

BRIGHTCART

Developer Guide

VERSION 1.0.0

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BrightCart is a React 18 single-page storefront built with Vite and styled with Tailwind CSS (`package.json`). This guide takes you from a fresh clone to a running dev server.

> **NOTE**
> Dependencies were not installed in the environment that generated this guide, so the npm commands below were read from `package.json` but not executed. They are marked `UNTESTED`.

Prerequisites

Tool	Version	Source
Node.js	20	`.nvmrc`, `node-version: 20` in `.github/workflows/ci.yml`
npm	bundled with Node 20	CI runs `npm ci`
Git	any recent	—

Clone to running

1. Clone the repository and enter it. The remote URL is not recorded in the repo `ASSUMPTION`.

```
git clone <repository-url> brightcart
cd brightcart
```

2. Create your environment file and fill in the values:

```
cp .env.example .env
```

Variable	Needed for
`VITE_API_URL`	Backend base URL used by `src/pages/Home.tsx`
`VITE_SENTRY_DSN`	Error reporting (`src/main.tsx`); empty disables it
`VITE_STRIPE_PUBLISHABLE_KEY`	Checkout (`src/pages/Checkout.tsx`)

3. Install dependencies `UNTESTED`:

```
npm install
```

4. Start the dev server `UNTESTED`:

```
npm run dev
```

Vite serves the app on `http://localhost:5173` by default `ASSUMPTION`. No port is set in the repo.

Project structure

```
brightcart/
├── index.html            # Vite entry; loads Poppins + Playfair Display
├── tailwind.config.js    # brand colours and font families
├── vercel.json           # SPA rewrite: every path serves /
├── public/               # logo.svg, favicon.svg
├── src/
│   ├── main.tsx          # React root + Sentry.init
│   ├── App.tsx           # routes: / and /checkout
│   ├── index.css         # Tailwind layers + CSS variables
│   ├── pages/            # Home.tsx, Checkout.tsx
│   ├── components/       # ProductCard.tsx
│   └── App.test.tsx      # Vitest + Testing Library
└── .github/workflows/ci.yml
```

Common workflows

Command	What it does	Tested
`npm run dev`	Start the Vite dev server	UNTESTED
`npm run build`	Type-check (`tsc -b`) and build to `dist/`	UNTESTED
`npm run preview`	Serve the production build locally	UNTESTED
`npm run lint`	ESLint over the repo	UNTESTED
`npm test`	Run Vitest once (`vitest run`)	UNTESTED

Add a page

1. Create `src/pages/<Name>.tsx`, following `src/pages/Home.tsx`.
2. Register a route in `src/App.tsx`:

```
<Route path="/orders" element={<Orders />} />
```

3. Production needs no server change: `vercel.json` already rewrites every path to `/`.

Use the brand styles

Use the Tailwind tokens from `tailwind.config.js` rather than raw hex values: `bg-brand`, `text-brand-700`, `font-display` (Playfair Display) and `font-sans` (Poppins).

Testing

```
npm test
```

`src/App.test.tsx` has 2 tests: the home page renders, and a placeholder for the checkout route. See Testing Strategy.

Debugging tips

- Vite only exposes variables prefixed with `VITE_` to the browser. A missing value reads as `undefined` rather than throwing, so check `.env` first when a feature silently fails.
- Sentry starts in `src/main.tsx` with `VITE_SENTRY_DSN`. Leave it empty locally to avoid sending development errors.
- `loadStripe` in `src/pages/Checkout.tsx` runs at module load. An invalid publishable key shows up in the browser console as soon as the checkout page is imported.

Related documents

Architecture · User Guide · Testing Strategy

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	Repository URL	assumption	no git remote in the repo	maintainer
A2	Dev server port 5173	assumption	Vite default, not configured	frontend lead
A3	npm commands work as listed	assumption	dependencies not installed during generation	frontend lead
Q1	Where does the backend behind `VITE_API_URL` live?	open question	not in this repo	engineering lead